

# Diagonalization

## Linear Algebra

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# Diagonalization Idea

## Definition (Similarity)

Suppose  $A$  is an  $n \times n$  matrix. We say that an  $n \times n$  matrix  $B$  is similar to  $A$  if there is an invertible  $n \times n$  matrix  $P$  so that

$$B = P^{-1}AP.$$

## Remark

Let  $T : V \rightarrow V$  be a linear transformation where  $V$  is a non-zero finite dimensional vector space. If  $A$  is the matrix of  $T$  with respect to a basis  $\mathcal{A}$  and if  $B$  is the matrix with respect to another basis  $\mathcal{B}$  of  $V$ , then  $B = P^{-1}AP$  where  $P$  is the change of basis matrix from the basis  $\mathcal{A}$  to  $\mathcal{B}$ .

## Definition (Diagonalizability)

A square matrix is said to be *diagonalizable* if it is similar to a diagonal matrix.

# Diagonalizability

## Theorem (Eigenvalues of Similar Matrices)

*Similar matrices have the same eigenvalues (including multiplicities).*

## Theorem

*An  $n \times n$  matrix  $A$  is similar to a diagonal matrix if and only if  $A$  is nondefective. In such a case, if  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$  denote  $n$  linearly independent eigenvectors of  $A$  and*

$$\boxed{P} = S = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n],$$



then

$$P^{-1}AP = S^{-1}AS = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n),$$

where  $\lambda_1, \lambda_2, \dots, \lambda_n$  are eigenvalues of  $A$  (not necessarily distinct) corresponding to the eigenvectors  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ .

# Diagonalizable Matrices: Counter Examples

## Example (Invertible but NOT Diagonalizable )

Let  $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Then  $A$  is invertible but not diagonalizable.

Example done

# Diagonalizable Matrices: Counter Examples

## Diagonalizability and Invertibility are Not Correlated

Among the following matrices, the first is diagonalizable and invertible, the second is diagonalizable but not invertible, the third is invertible but not diagonalizable, and the fourth is neither invertible nor diagonalizable, as the reader can verify:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

# Note/Remark on Diagonalizability

## Remark

An  $n \times n$  matrix with  $n$  distinct eigenvalues is diagonalizable.

## Note

If  $A$  has  $n$  linearly independent eigenvectors:

$$A = PDP^{-1}$$

where  $D$  is diagonal with eigenvalues.

Then:

$$A^k = PD^kP^{-1}$$

# Example: Diagonalization

## Example

Diagonalize the following matrix if possible:  $A = \begin{pmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{pmatrix}$ .

Step 1: Find the eigenvalues of  $A$ .

$$\det(A - \lambda I) = 0 \Rightarrow \begin{vmatrix} 3-\lambda & -2 & 4 \\ -2 & 6-\lambda & 2 \\ 4 & 2 & 3-\lambda \end{vmatrix} = 0.$$

$$\Rightarrow \lambda = 7, 7, -2.$$

$$\lambda_1 = 7$$

$$N(A - 7I_3)$$

$$E_{\{\lambda_1=7\}}$$

$$\begin{pmatrix} -4 & -2 & 4 \\ -2 & -1 & 2 \\ 4 & 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \boxed{-2x_1 - x_2 + 2x_3 = 0} \Rightarrow 2x_1 + x_2 - 2x_3 = 0.$$

$$x_1 = a \text{ and } x_3 = b \Rightarrow \boxed{x_2 = -2a + 2b}$$

$$\text{Another way: } x_1 = a, x_2 = b \Rightarrow x_3 = \dots$$

The sol<sup>n</sup>s will look like  $\begin{pmatrix} a \\ -2a+2b \end{pmatrix}, a, b \in \mathbb{R}$ .

$$= \begin{pmatrix} a \\ -2a \\ 0 \end{pmatrix} + \begin{pmatrix} b \\ 2b \\ b \end{pmatrix} = a \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$$



$$E_{\{\lambda_1=7\}} = \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \right\}$$

$$\boxed{\dim(E_{\{\lambda_1=7\}}) = 2}$$

$$\boxed{\lambda_2 = -2}$$

$$E_{\{\lambda_2=-2\}} = ?$$

and its dimension.

$N(A+2D)$

$$\begin{pmatrix} 5 & -2 & 4 \\ -2 & 8 & 2 \\ 4 & 2 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\boxed{\text{RREF}} \begin{pmatrix} 5 & -2 & 4 \\ -2 & 8 & 2 \\ 4 & 2 & 5 \end{pmatrix} \xrightarrow{R_{21}} \begin{pmatrix} -2 & 8 & 2 \\ 5 & -2 & 4 \\ 4 & 2 & 5 \end{pmatrix} \xrightarrow{-\frac{1}{2}R_1} \begin{pmatrix} 1 & -4 & -1 \\ 5 & -2 & 4 \\ 4 & 2 & 5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -4 & -1 \\ 5 & -2 & 4 \\ 4 & 2 & 5 \end{pmatrix} \xrightarrow[\substack{R_2 - 5R_1 \\ R_3 - 4R_1}]{\substack{R_2 - 5R_1 \\ R_3 - 4R_1}} \begin{pmatrix} 1 & -4 & -1 \\ 0 & 18 & 9 \\ 0 & 18 & 9 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & -4 & -1 \\ 0 & 18 & 9 \\ 0 & 0 & 0 \end{pmatrix}$$

$$x_1 + x_3 = 0$$

$$x_2 + \frac{1}{2}x_3 = 0$$

$$\Rightarrow x_1 = -x_3$$

$$\text{and } x_3 = -2x_2$$

$$\leftarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xleftarrow{R_1 + 4R_2} \begin{pmatrix} 1 & -4 & -1 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}$$

$\frac{1}{18}R_2$

The sol<sup>n</sup> will take the form:

$$\begin{pmatrix} x_1 \\ \frac{1}{2}x_1 \\ -x_1 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ \frac{1}{2} \\ -1 \end{pmatrix} = \frac{1}{2}x_1 \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

$$v_1 = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \text{ and } v_3 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 0 & 2 \\ -2 & 2 & 1 \\ 0 & 1 & -2 \end{pmatrix}$$

$$D = \begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & -2 \end{pmatrix} \\ = \text{diag}(7, 7, -2)$$

$$\boxed{A = P^{-1} \cancel{D} P}$$

Ans.

# Symmetric Matrices

## Definition (Symmetric Matrices/Operator)

A linear operator  $T \in L(V, V)$  on an inner product space is called symmetric if

$$\langle Tu, v \rangle = \langle u, Tv \rangle.$$

If  $T$  is represented by an  $n \times n$  square matrix  $A$  on  $V = \mathbb{R}^n$ , then a matrix is called symmetric if  $A^T = A$ .

## Theorem (Eigenvectors for Symmetric Matrices)

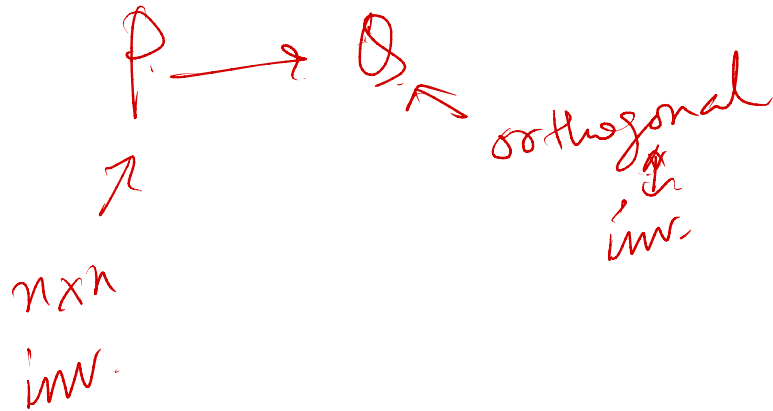
If  $A$  is symmetric, then two eigenvectors from different eigenspaces are orthogonal.

Proof: Let  $\lambda_1 \neq \lambda_2$   $v_1 \in E_{\lambda_1}$  and  $v_2 \in E_{\lambda_2}$

Aim:  $\langle v_1, v_2 \rangle = 0$

$$\begin{aligned} \langle \lambda_1 v_1, v_2 \rangle &= \lambda_1 \langle v_1, v_2 \rangle \\ &= \langle A v_1, v_2 \rangle = \langle v_1, A v_2 \rangle = \langle v_1, \lambda_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle \end{aligned}$$

Since  $\lambda_1 \neq \lambda_2 \Rightarrow \boxed{\langle v_1, v_2 \rangle = 0}$



## Special form of the matrix $P$

Therefore, normalizing the eigenvectors, the matrix  $P$  formed by the eigenvectors will consist of orthonormal columns, i.e.,  $P$  is an orthogonal.

## Definition

A matrix  $A$  is orthogonally diagonalizable if

$$A = PDP^{-1} = PDP^T$$

for some orthogonal matrix  $P$  and diagonal matrix  $D$ .

## Theorem

*An  $n \times n$  matrix  $A$  is symmetric if and only if it is orthogonally diagonalizable. In other words, we can say*

$$A = A^T \implies A = PDP^T.$$

# Example: Finding Orthogonal Diagonalizability

## Example

Diagonalize the following matrix if possible:  $A = \begin{pmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{pmatrix}$ .

$$b_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} \Rightarrow \|b_1\| = 1.$$

$$b_2' = v_2 - \frac{b_1 \cdot v_2}{b_1 \cdot b_1} b_1 = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \cdot 0 + 2 \cdot 2 \\ +1 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} - \frac{4}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$$

$$b_3 = \frac{1}{\|v_3\|} v_3 = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2/3 \\ 1/3 \\ -2/3 \end{pmatrix} \Rightarrow b_2 = \frac{1}{3\sqrt{5}} \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix}$$

$$b_1 = \begin{pmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \\ 0 \end{pmatrix} \quad b_2 = \begin{pmatrix} 4/3\sqrt{5} \\ 2/3\sqrt{5} \\ 5/3\sqrt{5} \end{pmatrix} \quad \text{and} \quad b_3 = \begin{pmatrix} 2/3 \\ 1/3 \\ -2/3 \end{pmatrix}$$

$$\boxed{b_1 \cdot b_2 = 0 = b_1 \cdot b_3 = b_2 \cdot b_3} \quad \|b_1\| = \|b_2\| = \|b_3\| = 1.$$

$$P = \begin{pmatrix} b_1 & b_2 & b_3 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{5} & 4/3\sqrt{5} & 2/3 \\ -2/\sqrt{5} & 2/3\sqrt{5} & 1/3 \\ 0 & 5/3\sqrt{5} & -2/3 \end{pmatrix}$$

Note:-  $\boxed{A = P^{-1} \cancel{A} P}$  —  $P$  is not unique.



$u, v$

$\downarrow$

$\lambda_1$

$(u+v)$

$$A(u+v) = Au + Av \\ = \lambda_1 u + \lambda_1 v$$

$$= \lambda_1 (u+v) \Rightarrow u+v \in E_{\lambda_1}$$

$$u, v \in E_{\lambda_1} \Rightarrow u+v \in E_{\lambda_1}$$

# Eigenvalues of Symmetric Matrices

## Theorem (Eigenvalues of Symmetric Matrices)

*If  $A$  is a symmetric  $n \times n$  matrix, then it has  $n$  real eigenvalues (counted with multiplicity) i.e., the characteristic polynomial  $p_A(\lambda)$  has  $n$  real roots (counted with repeated roots). Moreover,  $A$  has no defective eigenvalues. Eigenvectors correspond to distinct eigenvalues are orthogonal.*

# Linear Systems of ODEs and Its Connection to Linear Algebra

- A linear system of first-order ODEs can be written as

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x},$$

where:

- $\mathbf{x}(t) \in \mathbb{R}^n$
- $A \in M_n(\mathbb{R})$
- $A$  is a constant matrix

$$(x(t), y(t))$$

$$\dot{x}(t) = 2x + 3y$$

$$\dot{y}(t) = 3x - 4y$$

$$\mathbf{x}(t) \in \mathbb{R}^2$$

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$

$$\begin{pmatrix} \dot{x}(t) \\ \dot{y}(t) \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 3 & -4 \end{pmatrix} \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$$

## Example: $2 \times 2$ System

- We consider the following system

$$\begin{cases} x_1' = a_{11}x_1 + a_{12}x_2, \\ x_2' = a_{21}x_1 + a_{22}x_2. \end{cases}$$

- Which we can write in compact form:

$$\mathbf{x}' = A\mathbf{x}$$

where

$$\mathbf{x}' = \begin{pmatrix} x_1' \\ x_2' \end{pmatrix} \text{ and } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

# Why Diagonalize?

- Solving the above system of ODE directly can be complicated. If we can write:

$$A = PDP^{-1},$$

where:

- $D$  is diagonal
- $P$  contains eigenvectors of  $A$  (suppose  $A$  is diagonalizable).
- Then the system becomes much simpler.

# Change of Variables

- Let us introduce the following transformation:

$$\mathbf{x} = P\mathbf{y}.$$

$$\mathbf{x}' = P\mathbf{y}'$$

- By differentiating, we obtain

$$\mathbf{x}' = P\mathbf{y}'.$$

- By substituting the above into the system  $\mathbf{x}' = A\mathbf{x}$ , we get

$$P\mathbf{y}' = A P\mathbf{y}.$$

- Using the diagonalizability of  $A$  and the fact  $A = PDP^{-1}$ , we get

$$P\mathbf{y}' = P D \mathbf{y}.$$

$$PD = AP$$

- Finally, multiplying the above by  $P^{-1}$ , we get

$$\mathbf{y}' = D\mathbf{y}.$$

# Decoupled System

- We know that since  $A$  is diagonalizable we can write

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix},$$

where  $\lambda_1$  and  $\lambda_2$  are eigenvalues of  $A$ . Then

$$\begin{cases} y_1' = \lambda_1 y_1, \\ y_2' = \lambda_2 y_2. \end{cases}$$

*not decoupled*

- Each equation is independent.

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

# Solution of Decoupled System

- The solutions (transformed one) are

$$y_i(t) = c_i e^{\lambda_i t} \text{ for } i = 1, 2.$$

- Transforming back (to the original variables):

$$\mathbf{x}(t) = P\mathbf{y}(t).$$

- Thus, a solution to the original system of ODE is

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2.$$



# Thank You

Questions?